Tsinghua University

Hao Luo

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Primary Focus: DNA Data Storage

Postdoctoral Applications · August 2026

Additional Experience: Tumor Organoids and Assembloids; Engineered Living Materials

Microfluidics; Biofabrication; Automation

Academic Profile and Research Focus

Ph.D. researcher specializing in living DNA data storage. My primary research integrates information coding, engineered host and plasmid design, microencapsulation and preservation, random retrieval, regenerative rewriting, automation, and system-level hardware development. I established an engineered living memory microspheroid (ELMM) archival file system and subsequently developed a regenerative Living Disk–Drive architecture, advancing living DNA storage from file-level proof of concept toward searchable, regenerative, rewritable, and instrument-ready systems. Complementing this core research, I have contributed to patient-specific tumor organoid and assembloid models, with an emphasis on droplet-microfluidic fabrication and high-throughput drug screening.

- **Research output:** six peer-reviewed articles, including **two first-author papers** in ***Advanced Materials* (2025 and 2026)**, and **6 Chinese invention patent** filings, including three granted patents.
- **Interdisciplinary capability:** mechanical design, automation and control, Python/OpenCV, droplet microfluidics, synthetic biology, and end-to-end scientific instrument development, with additional experience in patient-derived tumor organoid and assembloid platforms.
- **Representative performance:** the ELMM platform achieved a theoretical retrieval throughput of 196.72 MB/s; at the volume of a household refrigerator (1.5 m^3), its theoretical storage capacity was estimated at approximately 260,700 PB (see the 2025 *Advanced Materials* paper and the Tsinghua Department of Mechanical Engineering report).

Education

Tsinghua University | Ph.D. Student in Mechanical Engineering, Institute of Forming Equipment and Automation — Sep. 2021 – Dec. 2026 (expected)

- Advisor: Prof. Zhuo Xiong; GPA: 3.8/4.0.
- Laboratory: Beijing Key Laboratory of Biofabrication and Regenerative Repair of Tissues and Organs.
- Dissertation research: living DNA information-storage systems, engineered storage units, microfluidics, and automated storage/retrieval instrumentation.

Tsinghua University | B.Eng. in Mechanical Engineering — Sep. 2017 – Aug. 2021

- GPA: 3.77/4.0; **ranked 3rd of 106 students (top 5%).**

Selected Research Contributions

Regenerative Living Disk–Drive System for Living DNA Storage | System architecture, process development, and instrument integration — Oct. 2024 – Dec. 2025

- Developed a system comprising a Living Disk, Optical Retriever, and Living Drive; integrated thermoresponsive release, bacterial amplification, re-encapsulation, and library replenishment into a retrieval–regeneration–use–replenishment cycle.
- Incorporated a CRISPR–Cas12a/λ-Red rewriting interface for coordinated replacement of information payloads, fluorescent indices, and selection markers. Published in *Advanced Materials* (2026, first author).

ELMM Archival File System ("Bacterial Bead Drive") | Full-stack process and algorithm development — Aug. 2023 – Oct. 2024

- Independently developed Python workflows for bidirectional conversion between digital files and DNA sequences; designed a droplet-microfluidic apparatus and established robust emulsification and encapsulation processes for approximately 50-μm bacterial microspheres, together with lyophilization protocols for room-temperature preservation.
- Used engineered bacterial fluorescence as a physical index and fluorescence-assisted sorting for random retrieval. Published in *Advanced Materials* (2025, first author) and resulted in related patent applications.

Patient-Specific Cancer Assembloid Manufacturing | Microfluidic platform development and process validation — Mar. 2021 – May 2022

- Designed microfluidic chips in AutoCAD and built the experimental platform for rapid encapsulation of limited patient-derived samples into microgels for high-throughput drug screening.
- Contributed to publications in *Nature Communications* and *Small Methods*, and to granted Chinese invention patent CN113583960B.

Chip-Based High-Throughput DNA Synthesizer | Automated prototype design and fabrication — Jul. 2022 – Aug. 2023

- Developed a DNA synthesizer prototype and a chip-coordinated coding strategy; demonstrated synthesis of a 13-nt DNA sequence and contributed to Chinese invention patent application CN117789788A.

Peer-Reviewed Publications

My name is shown in bold; † denotes first authorship. Article titles and DOI identifiers are clickable.

[1] **Luo, H.**†; Gao, J.; Huang, X.; Fang, Y.; Huang, T.; Xia, Y.; Yu, Z.; Cao, C.; Xiong, Z. Thermo-Responsive Living Microspheroids Enable a Regenerative Living Disk–Drive System for DNA Data Storage. *Advanced Materials* **38**(42), e73806 (2026). doi:10.1002/adma.73806. **First author**

[2] **Luo, H.**†; Huang, W.; He, Z.; Fang, Y.; Tian, Y.; Xiong, Z. Engineered Living Memory Microspheroid-Based Archival File System for Random Accessible In Vivo DNA Storage. *Advanced Materials* **37**(13), e2415358 (2025). doi:10.1002/adma.202415358. **First author**

[3] Zhang, Y.†; Hu, Q.†; Pei, Y.†; **Luo, H.**; et al. A Patient-Specific Lung Cancer Assembloid Model with Heterogeneous Tumor Microenvironments. *Nature Communications* **15**, 3382 (2024). doi:10.1038/s41467-024-47737-z.

[4] Xu, X.†; Gao, Y.; Dai, J.; Wang, Q.; Wang, Z.; Liang, W.; Zhang, Q.; Ma, W.; Liu, Z.; **Luo, H.**; et al. Gastric Cancer Assembloids Derived from Patient-Derived Xenografts: A Preclinical Model for Therapeutic Drug Screening. *Small Methods* **8**(9), e2400204 (2024). doi:10.1002/smtd.202400204.

[5] Ye, M.†; Shan, Y.; Lu, B.; **Luo, H.**; Li, B.; Zhang, Y.; Wang, Z.; et al. Creating a Semi-Opened Micro-Cavity Ovary Through Sacrificial Microspheres as an In Vitro Model for Discovering the Potential Effect of Ovarian Toxic Agents. *Bioactive Materials* **26**, 216–230 (2023). doi:10.1016/j.bioactmat.2023.02.029.

[6] Zhang, Y.†; Wang, Z.; Hu, Q.; **Luo, H.**; Lu, B.; Gao, Y.; Qiao, Z.; et al. 3D Bioprinted GelMA–Nanoclay Hydrogels Induce Colorectal Cancer Stem Cells Through Activating Wnt/β-Catenin Signaling. *Small* **18**(18), e2200364 (2022). doi:10.1002/smll.202200364.

Chinese Invention Patents

Patent titles are translated from the Chinese records; linked publication and grant numbers refer to the corresponding patent records.

[P1] Zhuo Xiong, Yanmei Zhang, Ting Zhang, and **Hao Luo**. Method and Device for Constructing Personalized Tumor Assembloids Based on Droplet Microfluidics. CN113583960B / ZL202110613983.6, granted in 2023.

[P2] Zhuo Xiong, **Hao Luo**, Liliang Ouyang, and Yu Yang. Methods and Apparatus for DNA-Based Information Storage and Readout. CN117789788A, published application (pending). **First-listed student inventor**

[P3] Zhuo Xiong, Ben Pei, Liliang Ouyang, and **Hao Luo**. Hierarchical Storage and Retrieval Method for DNA Data. CN115421669B / ZL202211217725.7, granted in 2025.

[P4] Min Ye, Ting Zhang, Zhuo Xiong, Bingchuan Lu, and **Hao Luo**. Ovulating Artificial Ovary Scaffold and Its Preparation Method and Application. CN114748690B / ZL202210248677.1, granted in 2022.

[P5] **Hao Luo** et al. Rewritable Engineered Living-DNA Information Storage Unit and Construction Method. Chinese invention patent application (2026). **First-listed student inventor**

[P6] **Hao Luo** et al. Hierarchical Living-DNA Data Storage System and Operation Method. Chinese invention patent application (2026). **First-listed student inventor**

Additional Research and Engineering Experience

NSFC General Program Proposal | Primary drafting and revision contributor; advisor listed as proposed principal investigator — Dec. 2025 – Mar. 2026

- Contributed to research framing, technical-route development, and proposal preparation.

Dot-Matrix Recognition Algorithm | Core algorithm development, system packaging, and debugging — Jan. 2025 – Jun. 2025

- Developed image preprocessing, feature extraction, and dot-matrix recognition pipelines in Python and OpenCV; packaged the core code as a DLL software module.

Dynamic-Focusing Galvanometer and Embedded Servo Control | Core developer, National Undergraduate Innovation Program — Sep. 2020 – May 2021

- Fabricated the mechanical system, refined component and control designs, and integrated an STM32 microcontroller with sensors and actuators for high-precision real-time control.

Duneng Vision: Computer-Vision-Based Factory Monitoring | Project member, Tsinghua X-lab
Jul. 2021 – Jan. 2022

- Evaluated computer-vision methods for factory 6S management, congestion monitoring, and personnel/cargo recognition; related algorithms were piloted in a JD Logistics setting.

Teaching Experience

Teaching Assistant, Biomaterials Engineering and Devices | Tsinghua University
Jul. 2021 – Aug. 2026

- Supported lectures and laboratory instruction, improved teaching workflows, and received the Tsinghua University Outstanding Teaching Assistant Award (2022; ranked in the top 5% university-wide in course evaluations).
- Contributed to an AI-enabled curriculum reform initiative and the "Biomanufacturing Brain" teaching platform.

Teaching Assistant, Physics Laboratory B | Tsinghua University
Sep. 2025 – Jun. 2026

- Explained experimental principles, assembled laboratory setups, and supervised undergraduate experiments.

Teaching Assistant, Fundamental Physics | Tsinghua University
Sep. 2025 – Jun. 2026

- Supported course delivery, problem-solving sessions, and student feedback.

Conference Presentations and Participation

[T1] **Hao Luo**. "Thermo-Responsive Living Microspheroids Enable a Regenerative Living Disk–Drive System for DNA Data Storage." The 6th International Conference on Biomaterials, Bio-Design and Manufacturing (BDMC 2026), The Chinese University of Hong Kong, Hong Kong, China, 22–24 July 2026. **Oral presentation**

[T2] **Hao Luo**. "Engineered Living Memory Microspheroid-Based Archival File System for Random Accessible In Vivo DNA Storage." The 31st International Conference on Computational & Experimental Engineering and Sciences (ICCES 2025), Changsha, China, 25–29 May 2025. **Oral presentation**

[T3] **Hao Luo**. "Research on the Fabrication of Cancer Assembloids Based on Droplet Microfluidics." 2023 National Conference on Biomanufacturing Engineering and International Symposium on Biomanufacturing (ACBD–ISBM 2023), Beijing, China, 17–19 March 2023. **Oral presentation**

[T4] Beijing–Tianjin–Hebei Frontier Workshop on DNA Storage, Wuqing, Tianjin, China, 29–30 April 2023. **Participant**

Selected Media Coverage

[M1] Sina News: Tsinghua Team Develops a Regenerative Living DNA Disk–Drive Storage System, 3 July 2026.

[M2] Tsinghua University News: Xiong Research Group Develops a Regenerative Living DNA Disk–Drive Storage System, 1 July 2026.

[M3] Tencent News: Tsinghua Develops a "Bacterial Bead Drive"—How Can a Refrigerator-Sized Device Store 260 Million TB, and Could It Be Used in Data Centers?, 9 March 2025. **Featured Q&A**

[M4] Tencent News: Tsinghua University Breaks the DNA Storage Bottleneck—Theoretical Retrieval Speed Reaches 196.72 MB/s, 1 March 2025.

[M5] Sina Finance: Tsinghua University Breaks the DNA Storage Bottleneck—Approximately 267 Million TB Can Be Stored in 1.5 m^3, 1 March 2025.

[M6] Phoenix News: Tsinghua University Develops a "Bacterial Bead Drive"—Breaking the DNA Storage Bottleneck, 1 March 2025.

[M7] Tsinghua University News: Xiong Research Group Develops a "Bacterial Bead Drive" for Rapid Random Access in DNA Data Storage, 27 February 2025.

[M8] Nanowerk Spotlight: Bacteria Encapsulated in Microspheres Create Living, Searchable DNA Data Storage System, 25 February 2025.

Honors and Awards

2022	Second-Class Comprehensive Excellence Award, Tsinghua University; Outstanding Teaching Assistant Award, Tsinghua University (top 5% university-wide in course evaluations)
2020	Energy Science Scholarship, Tsinghua University (Comprehensive Excellence Category)
2018, 2019	Academic Excellence Award, Tsinghua University (two consecutive years)

Leadership and Organizational Experience

Graduate Student Cohort 213, Department of Mechanical Engineering, Tsinghua University | Cohort Representative and Student Organization Lead — Sep. 2021 – Jan. 2023

- Coordinated cohort activities, student engagement initiatives, and team collaboration; the cohort was recognized as a Tsinghua University Class-A Student Organization.

Mechanical Engineering Class 72 Youth League Branch, Tsinghua University | Youth League Branch Secretary — Sep. 2020 – Jun. 2021

- Coordinated class activities and academic-culture initiatives, and led the class to receive the Tsinghua University Excellent Academic Culture Class designation.

Mechanical Engineering Class 72 Party Theory Study Group, Tsinghua University | Group Leader — Sep. 2019 – Sep. 2020

- Led political-theory study and educational activities, strengthening class cohesion and values-based guidance.

Student Learning and Development Department, Tsinghua Mechanical Engineering | Staff Member — Jan. 2018 – Dec. 2018

- Assisted in organizing academic events and peer-learning activities, supporting student development and a strong academic culture within the department.

Industry and Internship Experience

Guangdong Shunde Innovation Design Institute | R&D Assistant, R&D Department — Jun. 2023 – Aug. 2023

- Proposed a rapid-forming process for artificial blood vessels based on GelMA–PVA materials and completed the material-preparation and forming-process design.

Shenzhen Silver Basis Technology Co., Ltd. | Structural Design Intern; Team Leader and Project Lead — Jul. 2020 – Aug. 2020

- Designed an axial-cam compression rotary engine; used SolidWorks for three-dimensional modeling and assembly, and analyzed force, power, water-cooling, and sealing schemes.
- Used ANSYS to evaluate stress and thermal performance and AutoCAD to develop machining and assembly workflows; the design resulted in one Chinese patent application.

Technical Skills

Biology / microfluidics	Plasmid and host design, engineered bacterial culture, droplet microfluidics, hydrogel-microsphere encapsulation, lyophilization, and organoid and assembloid manufacturing
Mechanics / control	SolidWorks, AutoCAD, ANSYS, mechanical structure and transmission design, CNC machining, 3D printing, STM32/Arduino control, and scientific instrument and system integration
Computing / software	Python, OpenCV, MATLAB, C/C++, Keil, Arduino IDE, DNA information coding/decoding, image processing, dot-matrix recognition, and DLL software packaging